

Autovol Automated Framing Line

Machine Breakdown & Steel-Frame Adaptation Notes

This document is a supplementary reference for the New Eden digital twin. It is a deeper, machine-by-machine companion to *New-Eden-Digital-Twin-References.pdf* (already in this folder), which briefly cites the category of automated cold-formed-steel framing tables New Eden's Walls lane is modeled on "in the spirit of Autovol-style automated framing factories." This document unpacks that single reference line into its constituent stations, sourced from a direct visual review of a real promotional video, plus supplementary web research, and pairs every timber-line station with a plain-language note on what New Eden — a steel-only, cold-formed light-gauge steel modular factory — would need instead.

Source video: "Transforming Construction with Autovol" — published on ABB Robotics' official YouTube channel (youtu.be/9rkLPEWwFv4, runtime 3:35). The findings below come from direct visual inspection of scene-change frames extracted from this exact video, including its burned-in interview captions, cross-checked against independent web sources for company facts, dates, and product lines. Where a detail could not be independently confirmed beyond what is visible on screen, that is stated plainly rather than guessed.

Who is shown in the video: **Autovol, Inc.** is an automated volumetric modular construction company based in Nampa, Idaho. Its 400,000-sq-ft factory (opened 2020, ~\$100M+ investment) uses robotic systems to build fully enclosed six-sided modules — walls, floors, ceilings, and MEP — to roughly 80% completion before shipping them to the building site. As of 2023 the company had produced 1,500+ modules across 13+ multifamily projects, and reported \$125M revenue in 2025. **House of Design (also referred to as "House of Design Robotics")** is a Nampa, Idaho robotics systems-integration company, founded by Shane Dittrich, that designed, engineered, and built Autovol's automation cells; House of Design is an ABB Robotics "Value Provider" and also integrates Cognex machine vision. **ABB Robotics** supplied the industrial robot hardware — Autovol's factory runs 53 ABB six-axis industrial robots handling framing, fastening, sheathing, and material movement.

Framing note: Autovol's line is a **timber** platform — dimensional lumber studs, OSB/gypsum sheathing, nail/staple/screw fastening. New Eden is **steel-only**: roll-formed cold-formed steel (CFS) C-studs and tracks, self-drilling screw or resistance-spot-weld fastening, non-combustible sheathing. No station below is a claim that New Eden will run this timber process as-is — every entry ends with an explicit steel-frame equivalent, and none of this implies any compliance certification (see disclaimer, final page).

Machine-by-Machine Breakdown

Fifteen distinct machines/stations were identified across the extracted frames. Each entry lists what it does in Autovol's timber line, the real manufacturer/brand where identifiable, the source for that identification, and New Eden's steel-frame equivalent.

1. ABB 6-axis industrial robots (general pick-and-place)

Function: Multiple white ABB articulated robots pick and place framing members, position panels, and palletize parts throughout the line.

Real brand / manufacturer: ABB Robotics. Autovol's factory runs 53 ABB six-axis industrial robots in total, across all stations (framing, sheathing, fastening, material handling).

Verification note: No official Autovol-specific ABB model number is publicly published; the IRB 6700/6640/4600 families are named here only as ABB's general-purpose payload classes that suit this kind of work — not confirmed as Autovol's exact model.

Source: AEC Business, "Autovol: An Automated Volumetric Modular Factory Taking Shape" — aec-business.com

Source: Building Design + Construction, "Rick Murdock's dream multifamily housing factory" — bdcnetwork.com

Source: ABB Group press release, "ABB Robotics advances construction industry automation," 2021-05-20 — new.abb.com/news/detail/78359

New Eden steel-frame equivalent: New Eden's digital twin already models an ABB IRB 8700-550/4.2 for the truss-welding cell (per the existing References PDF), spec-accurate to ABB's published datasheet. For general pick-and-place of roll-formed steel C-studs and tracks, the equivalent role is filled by lighter/medium-payload ABB industrial robots (e.g., the IRB 6700 or IRB 4600 families by published payload class) fitted with grippers shaped for C-section steel profiles rather than rectangular lumber — steel members are thinner and sharper-edged, so gripper jaw geometry and clamping force must change, not just scale down.

2. ABB robots with red quick-change fastening tooling

Function: ABB robots fitted with a custom red aluminum end-of-arm tool (cast with the ABB logo) that drives fasteners into framing/sheathing.

Real brand / manufacturer: ABB robot arm confirmed; the red tool itself appears to be a custom/OEM-integrator-built end effector (consistent with House of Design's role building custom tooling around ABB's quick-change tool interface). No public spec sheet for this specific tool was found.

Verification note: *Manufacturer of the physical red tool body could not be confirmed from public sources; treat as unverified.*

Source: Technology Magazine, "Automation beyond imagination," 2021-07-19 (House of Design/Autovol case) — technologymagazine.com

Source: Direct frame inspection of source video (ABB Robotics YouTube channel)

New Eden steel-frame equivalent: New Eden's steel line replaces nail/staple/screw-gun tooling of this kind with a self-drilling, self-tapping TEK-screw driving head (the standard cold-formed-steel fastening method) or a robotic resistance-spot-weld gun for stud-to-track connections. Screws for CFS framing seat differently than for wood — no pilot splitting risk, but torque/depth control against thin galvanized steel is critical to avoid stripping or over-drive, which is a different control problem than wood fastening.

3. Multi-spindle automated fastening ("nail-gang") head

Function: A roughly 6-gang red fastening head with black nose pieces, mounted on an overhead gantry, drives multiple fasteners simultaneously across a wall or floor panel's width.

Real brand / manufacturer: Brand could not be identified from the visible frames or from web research; gang-nailer / multi-spindle fastening heads are a known equipment category in automated timber-frame manufacturing but no matching product page was found to confirm this specific unit's maker.

Verification note: *Brand/manufacturer of this specific fastening head is unverified — flagged rather than guessed.*

Source: Direct frame inspection of source video (ABB Robotics YouTube channel) — no independent manufacturer match found

New Eden steel-frame equivalent: This is the single station with the clearest wood-vs-steel process split. Nail-gang heads exist because nails can be driven fast and simultaneously into softwood. Cold-formed steel framing does not use nails at all — connections are self-drilling screws (per cold-formed steel construction practice) or spot welds. New Eden's equivalent is a multi-spindle self-drilling screw-gang head, or a multi-electrode spot-weld array, sized to New Eden's stud spacing rather than a lumber module's on-center spacing.

4. Güdel overhead linear gantry ("FLOOR GANTRY")

Function: An overhead gantry rail system carrying robots along the length of the framing tables; on-screen label reads "FLOOR GANTRY." The GÜDEL logo is clearly legible in the source frames.

Real brand / manufacturer: Güdel Group AG (Switzerland) — a real, established manufacturer of linear-motion modules, robot track systems, and gantry robots (TrackMotion Floor/Overhead/Vertical product lines; multi-axis gantries used across automotive, aerospace, and heavy-industrial automation). Güdel has separately published its own gantry-robot work in automated timber processing (a WoodFlex 56 multi-function gantry performing cutting, nailing, and panel handling for a wood-construction customer), which corroborates why a company like Güdel supplies this category of machine to automated framing lines generally.

Verification note: *No public case study confirms Güdel officially names Autovol as a customer; identification here is from the visible on-screen logo in the source video, corroborated by Güdel's general product line.*

Source: [Güdel corporate site — gudel.com/products/linear-tracks-for-robots](https://gudel.com/products/linear-tracks-for-robots)

Source: Güdel news, "Fascinating process expertise at Güdel" (WoodFlex 56 gantry, wood processing) — gudel.com/news/the+future+of+modular+construction+today_n7831

Source: Automate.org, "Güdel Showcases Broadest Product Line of Robot Tracks and Gantries" — automate.org

New Eden steel-frame equivalent: Same category of machine transfers directly: New Eden's Walls/Floors lanes would use an equivalent overhead or floor-mounted linear gantry (Güdel TrackMotion-class or equivalent) to carry robots along steel framing tables. The gantry itself is largely material-agnostic — the adaptation is in table length/width and end-effector payload sized to New Eden's steel member dimensions, not the gantry's linear-motion hardware.

5. Vacuum/suction-cup sheathing lifter

Function: A yellow suction-cup array end-effector on an overhead beam lifts full sheathing sheets (OSB) onto the framing table.

Real brand / manufacturer: Vacuum lifting for sheet materials is a standard, widely available industrial technology (suppliers include companies such as Schmalz, Anver, and Vacuworx); no specific brand is identifiable from the visible frames.

Verification note: *Specific manufacturer of the visible suction-cup array is unverified.*

Source: [Direct frame inspection of source video \(ABB Robotics YouTube channel\)](#) — [brand not independently confirmed](#)

New Eden steel-frame equivalent: Vacuum lifting works equally well, arguably better, for New Eden: sheet steel decking/sheathing and flat non-combustible board are both good vacuum-lift candidates, and steel sheet also opens the option of magnetic lifting as an alternative/supplement, which timber sheathing cannot use. Cup layout and lift capacity would be resized for the specific sheet product New Eden specifies.

6. ABB dual-arm collaborative robot (YuMi/GoFa-style)

Function: A white/black ABB collaborative dual-arm robot shown in a smaller precision demo cell, used for finer manipulation than the large floor robots.

Real brand / manufacturer: ABB Robotics — ABB's YuMi (IRB 14000) is ABB's dual-arm collaborative robot product line; ABB also makes single-arm GoFa cobots. The unit shown matches ABB's published cobot family generally; exact model not confirmed from the visible frames.

Source: [ABB collaborative robots product line — cobots.robotics.abb.com](https://cobots.robotics.abb.com)

Source: Technology Magazine, "Automation beyond imagination" (references ABB's YuMi/GoFa/SWIFTI cobot lines) — technologymagazine.com

New Eden steel-frame equivalent: New Eden's equivalent role is precision QA/small-parts work on steel components — e.g., installing small brackets/clips, verifying punched connection-hole alignment, or handling electrical/MEP inserts before a module is closed up — tasks suited to a cobot's lower payload and closer human collaboration rather than the heavy structural robots.

7. "AUTO-ROLL" branded powered roller-conveyor

Function: A powered roller-conveyor system, branded "AUTO-ROLL" on screen, moves panels/modules along the line.

Real brand / manufacturer: Could not be independently verified. No manufacturer or product line named "AUTO-ROLL" was found in web research matching this application; it may be a smaller/regional conveyor brand, a private-label name, or a name specific to Autovol's own line.

Verification note: *Brand "AUTO-ROLL" is unverified — flagged rather than attributed to a guessed manufacturer.*

Source: [Direct frame inspection of source video \(ABB Robotics YouTube channel\)](#) — no independent manufacturer match found

New Eden steel-frame equivalent: Functionally, powered roller conveyors are a mature, material-agnostic technology (multiple manufacturers supply heavy-duty industrial roller conveyor). New Eden's version would need a higher rated load capacity line, since steel-framed modules generally weigh more per unit volume than timber-framed modules of the same size — this is a sizing/rating adaptation, not a different machine category.

8. Blue steel framing/assembly jig tables

Function: Blue steel fixture tables where wall and floor timber frames are squared, held, and fastened.

Real brand / manufacturer: This is the general category of automated/semi-automated framing table used across light-gauge and timber framing manufacturing; no single named manufacturer is visible on screen. This is the same category New Eden's existing References PDF already cites in the spirit of "Autovol-style automated framing factories," alongside FRAMECAD's published automated cold-formed-steel framing systems.

Source: [BuildSteel.org, "Robotic Automation in Cold-Formed Steel Construction"](#) — [buildsteel.org](#)

Source: [FRAMECAD — automated steel framing systems](#) — [framecad.com](#)

Source: [AAC Steel, "Precision Manufacturing for Cold-Formed Steel Framing"](#) — [aacsteel.com](#)

New Eden steel-frame equivalent: This is the one station where New Eden's equivalent is the *same underlying concept*, not an adaptation: New Eden's Walls/Floors lanes already use squaring/holding jig tables in this category, just built to hold roll-formed steel C-studs and tracks (AISI S240 profiles) instead of dimensional lumber — different pin/clamp spacing and hold-down force for thin galvanized steel versus timber.

9. AutoVol-branded self-propelled transport cart

Function: A red, self-propelled transport cart with a lift/tilt arm, branded "AutoVol," moves finished panels/modules across the factory floor.

Real brand / manufacturer: Appears to be Autovol's own proprietary/custom-built factory equipment (branded with the company's own name rather than a third-party equipment maker). No public spec sheet found — consistent with this being in-house material-handling equipment rather than an off-the-shelf product.

Verification note: *No manufacturer beyond Autovol's own branding could be confirmed; treat as proprietary/uncited equipment.*

Source: [Direct frame inspection of source video \(ABB Robotics YouTube channel\)](#) — no third-party manufacturer identified

New Eden steel-frame equivalent: Same functional role — a self-propelled cart moving finished assemblies between stations — but structurally rated for New Eden's steel-framed panel/module dead loads, which are typically

heavier than an equivalent timber-framed assembly.

10. Yellow tripod laser measurement/leveling instrument

Function: A yellow tripod-mounted laser instrument used for alignment and QA verification on the line.

Real brand / manufacturer: Laser levels/total-station instruments are a standard commercial product category (major suppliers include Leica Geosystems, Topcon, and Trimble); the specific brand is not legible in the available frames.

Verification note: *Specific instrument brand is unverified.*

Source: [Direct frame inspection of source video \(ABB Robotics YouTube channel\)](#) — brand not independently confirmed

New Eden steel-frame equivalent: Same QA function transfers directly — laser alignment/verification instruments are material-agnostic. The meaningful difference is what tolerance band New Eden checks against: cold-formed steel framing has different dimensional tolerance expectations than timber framing, governed by structural steel design standards (see AS/NZS 4600 reference in the existing References PDF) rather than timber-framing tolerances — this document does not claim New Eden's build meets any such standard, only that a real build would be checked against it.

11. Precision marking/cutting station (partial brand "...LTRONICS")

Function: A precision marking/cutting station appears in the line; the visible brand text is partial, ending "...LTRONICS."

Real brand / manufacturer: Could not be confirmed. Web research into CNC timber-cutting/marketing equipment makers (e.g., Hundegger, Weinig) did not surface a matching name ending in "...LTRONICS." This is flagged as unverified rather than guessed, per this project's sourcing rules.

Verification note: *Brand ending "...LTRONICS" could not be identified — explicitly unverified.*

Source: [Direct frame inspection of source video \(ABB Robotics YouTube channel\)](#) — no confirmed manufacturer match

New Eden steel-frame equivalent: New Eden does not need a separate freestanding timber marking/cutting station in the same place in the line: New Eden's roll-forming line (modeled on Sunway-style floor-deck roll formers, per the existing References PDF) already integrates in-line punching of connection holes and part labelling as coil steel is roll-formed into studs and tracks — that upstream station absorbs the marking/cutting function this timber-line station performs separately.

12. House of Design Robotics CAD/CAM engineering workstation

Function: A dual-monitor engineering workstation showing a 3D robot/building model, used to program the ABB robots directly from the building's 3D design.

Real brand / manufacturer: House of Design (Nampa, Idaho) — the real ABB Robotics Value Provider / systems integrator that engineered Autovol's automation. House of Design engineers use ABB's own RobotStudio offline programming/simulation software to program the robot cells from the building's 3D data.

Source: [House of Design — thehouseofdesign.com/about-us/autovol](#)

Source: [Building Design + Construction, "Rick Murdock's dream multifamily housing factory" \(confirms RobotStudio use\)](#) — [bdcnetwork.com](#)

Source: [Idaho Commerce, "Company Spotlight: House of Design"](#) — [commerce.idaho.gov/blog/house-of-design](#)

Source: [Modular Building Institute, "Robots are the Future of Modular Construction" \(Shane Dittrich, House of Design founder\)](#) — [modular.org](#)

New Eden steel-frame equivalent: Same pipeline concept applies to New Eden: a CAD/CAM workstation that converts New Eden's steel-frame BIM model (stud/track layout, punch/connection-hole positions, roll-form nesting data) directly into robot programs, rather than timber framing plans. The engineering discipline required is cold-formed-steel specific (AISI S240 profile geometry, screw/weld schedules) rather than timber framing-specific.

13. Automated fastener supply ("SCREW" bulk boxes)

Function: Bulk boxes labeled "SCREW" feed the robotic fastening tools with fasteners.

Real brand / manufacturer: Generic industrial fastener supply packaging; no brand identifiable. Notably, the labeling confirms Autovol's line uses screws (not only nails) for at least some fastening passes, consistent with the video's own captioned claim that the process can "nail, glue, or screw all the parts."

Source: [Direct frame inspection of source video \(ABB Robotics YouTube channel\)](#)

New Eden steel-frame equivalent: New Eden's steel framing uses self-drilling/self-tapping TEK screws as the primary mechanical fastener for stud-to-track connections (no nails, no glue, per standard cold-formed steel construction practice), so this bulk-supply concept transfers directly — just with a steel-specific screw type/coating (corrosion protection matters more on exposed galvanized steel connections) feeding the screw-gang head described in item 3.

14. Doosan-branded mobile material-handling equipment

Function: A yellow scissor-lift/forklift-type mobile unit, visibly branded Doosan, used for material handling around the factory floor.

Real brand / manufacturer: Doosan (Doosan Industrial Vehicle / Doosan Bobcat) is a real, major manufacturer of forklifts, scissor lifts, and material-handling equipment. The Doosan brand is legible in the source frames; no specific model number could be confirmed from the visible footage, and no Autovol/Doosan case study or partnership announcement was found in web research beyond this visual identification.

Verification note: *Model number not confirmed; brand is a direct visual read from the source video only.*

Source: [Direct frame inspection of source video \(ABB Robotics YouTube channel\)](#) — brand visually confirmed on equipment; no published Autovol/Doosan case study found to corroborate model/spec

New Eden steel-frame equivalent: Same equipment category transfers directly for coil-steel and finished-module handling around New Eden's floor, sized to New Eden's heavier steel-framed loads and to the specific dimensions of coil steel stock (coils are handled differently than palletized lumber — typically via coil cars/upenders rather than forks).

15. End-of-line shrink-wrap and dispatch

Function: A finished module is shrink-wrapped and loaded onto a dolly/flatbed for dispatch to the job site.

Real brand / manufacturer: Standard industrial packaging/logistics practice; no specific equipment brand identifiable from the frames.

Source: [Direct frame inspection of source video \(ABB Robotics YouTube channel\)](#)

New Eden steel-frame equivalent: Same finished-module wrap-and-ship step applies to New Eden. One material-specific note worth flagging for a real build (not a claim, just a planning consideration): cut/punched edges on galvanized steel framing can be more corrosion-sensitive than a sealed timber assembly during transport and staging, which a real engineering review would need to address — this document does not claim New Eden has solved or certified against that.

The Systems-Integrator Model: Autovol / House of Design / ABB

Autovol's automation is not built by a single company. It is a three-tier structure, and the video's own burned-in captions describe it directly:

"Our partnership with ABB Robotics is formally called a value provider. We use our engineering services to provide their robotics products to our customers... so if you're a manufacturer and you want to automate a solution, you can use an ABB robot but you're going to need a House Of Design Robotics company to provide all the value-added engineering services."

— Autovol / House of Design Robotics promotional video, ABB Robotics YouTube channel

"ABB Robotics was introduced to us through House of Design Robotics... one of their providers... with an intricate knowledge of automation and technology, and so they supplied the machinery & the technology for us to be able to do the things we do with automation and robotics."

— Autovol / House of Design Robotics promotional video, ABB Robotics YouTube channel

The three roles, as corroborated by independent sources:

Role	Company	What they actually do
Operating company / customer	Autovol, Inc.	Defines the product (volumetric modules), owns the factory and process, operates the line day-to-day.
Systems integrator / "Value Provider"	House of Design	Does the mechanical/electrical engineering, programming (via ABB RobotStudio), custom tooling, assembly, and testing that turns generic ABB robots into a working, application-specific factory line.
Robot/component OEM	ABB Robotics	Manufactures and supplies the underlying robot hardware, controllers, and software platform (RobotStudio, OmniCore) that the integrator builds on top of.

House of Design's own materials describe this concretely: "House of Design develops the process, does all of the mechanical engineering, electrical engineering, programming, assembly, tests in their facility and then packages it up and ships it to customers." ABB, for its part, frames House of Design and similar integrators as recipients of its formal "Value Provider Solution Award" program, which recognizes integrator partners (a Canadian integrator, Brave Control Solutions, won the same award in 2021 for a construction-industry automation process).

Sources: [House of Design capabilities brief \(Autovol brochure, autovol.com\)](#); [Idaho Commerce, "Company Spotlight: House of Design" \(commerce.idaho.gov\)](#); [Technology Magazine, "Automation beyond imagination" \(technologymagazine.com\)](#); [Modular Building Institute, "Robots are the Future of Modular Construction" \(modular.org\)](#).

Why this matters for New Eden

New Eden's digital twin currently visualizes an integrated automation stack in one model, but a real build would very likely need this same three-tier structure rather than one company doing everything in-house. The steel-specific equivalent of House of Design's role — the systems integrator that turns generic industrial robots into a working, application-specific line — would need **cold-formed steel framing domain expertise specifically**, not just general robotics integration experience: grippers for C-stud/track profiles, self-drilling screw or spot-weld end-of-arm tooling, and jig-table designs built around AISI S240 stud/track geometry are a different engineering problem than the timber tooling House of Design built for Autovol. Companies already publishing in this specific space include FRAMECAD, whose product line is automated cold-formed-steel framing systems ([framecad.com](#)), and the broader cold-formed steel robotics work catalogued by BuildSteel.org ("Robotic Automation in Cold-Formed Steel Construction,"

buildsteel.org) — both already cited in New Eden's existing References PDF as the category New Eden's Walls lane draws on. This document does not claim New Eden has selected, contracted, or partnered with any integrator, OEM, or vendor named anywhere in this document — it is a reference for planning purposes only.

Disclaimer

This document is a conceptual research/reference deliverable for the New Eden digital twin project. It does not claim that New Eden has purchased, licensed, partnered with, or been certified by any company, product, or standard named above — including AutoVol, House of Design, ABB Robotics, Güdel, Doosan, FRAMECAD, or any standards body. Machine identifications are grounded in direct visual inspection of a real, publicly available promotional video plus independently sourced web research; anything that could not be independently verified is explicitly flagged as unverified rather than presented as fact. New Eden itself is a visualization/planning project, not a built factory, and does not track or claim compliance with any real-world engineering, safety, or building-code certification. Before any physical construction based on this concept, New Eden would need formal review and sign-off from licensed structural, mechanical, electrical, and safety engineers.

Compiled for the New Eden project — companion to [digital-twin/New-Eden-Digital-Twin-References.pdf](#)